

Example: Use Type II transformation (unique solution) to solve Type I transformation

To find $f_W(w)$ given $W = g(X, Y)$ and $f_{XY}(x, y)$

► Let $Z = Y$ and use previous approach to find $f_{W,Z}(w, z)$

► Find marginal $f_W(w) = \int f_{W,Z}(w, z) dz$

If $W = g(X, Y) = aX + bY$, to find $f_W(w)$, let $Z = Y$, then

$$\begin{bmatrix} w \\ z \end{bmatrix} = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = A \begin{bmatrix} x \\ y \end{bmatrix}, \quad \text{where } A = \begin{bmatrix} a & b \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad A^{-1} = \frac{1}{a} \begin{bmatrix} 1 & -b \\ 0 & a \end{bmatrix} = \begin{bmatrix} 1/a & -b/a \\ 0 & 1 \end{bmatrix}$$

Thus, $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1/a & -b/a \\ 0 & 1 \end{bmatrix} \begin{bmatrix} w \\ z \end{bmatrix}$, resulting in $x = \frac{w - bz}{a}$ and $y = z$

Since $|\det(A^{-1})| = \frac{1}{|a|}$, then $f_{W,Z}(w, z) = \frac{1}{|a|} f_{XY}\left(\frac{1}{a}(w - bz), z\right)$, and

$$f_W(w) = \int \frac{1}{|a|} f_{XY}\left(\frac{1}{a}(w - bz), z\right) dz$$

Transformation Type II without unique solution (pages 9-14)

Example: Let X and Y be noisy voltage measurements with known joint CDF $F_{XY}(x, y)$

If only interested in maximum and minimum values between the two measurements then two new RVs: $W = \min(X, Y)$ and $Z = \max(X, Y)$

To find joint CDF of W and Z ,

$$\begin{aligned} F_{WZ}(w, z) &= \Pr(W \leq w, Z \leq z) \\ &= \Pr(\min(X, Y) \leq w, \max(X, Y) \leq z) = \Pr((x, y) \in R_w, (x, y) \in R_z) \end{aligned}$$

Need to find points (x, y) in the **intersection** between two regions

R_w : (x, y) such that minimum between x and y is less than w

$\Rightarrow$ either $x \leq w$ or $y \leq w$ or both (union)

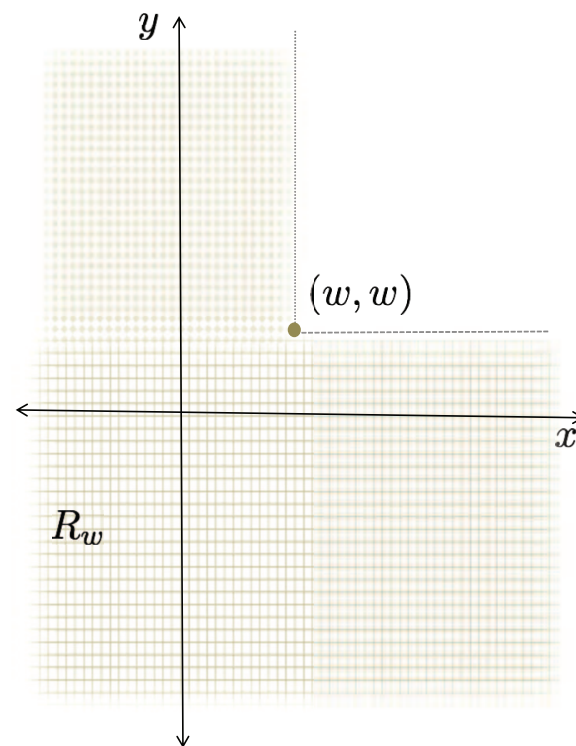
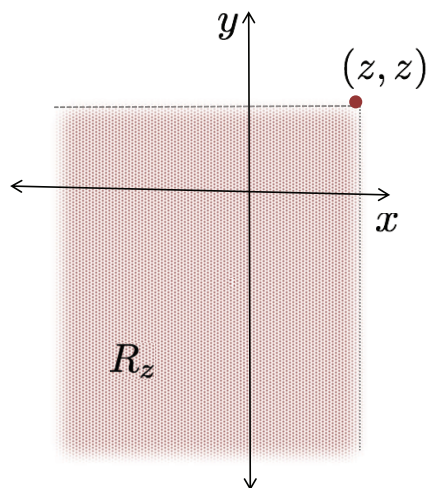
R_z : (x, y) such that maximum between x and y is less than z

$\Rightarrow$ both $x \leq z$ and $y \leq z$ (intersection)

To find (x, y) points in the intersection between regions

► R_w : $\{(x, y) : \min(\mathbf{x}, \mathbf{y}) \leq \mathbf{w} : \mathbf{x} \leq \mathbf{w} \text{ or } \mathbf{y} \leq \mathbf{w} \text{ or both}\}$

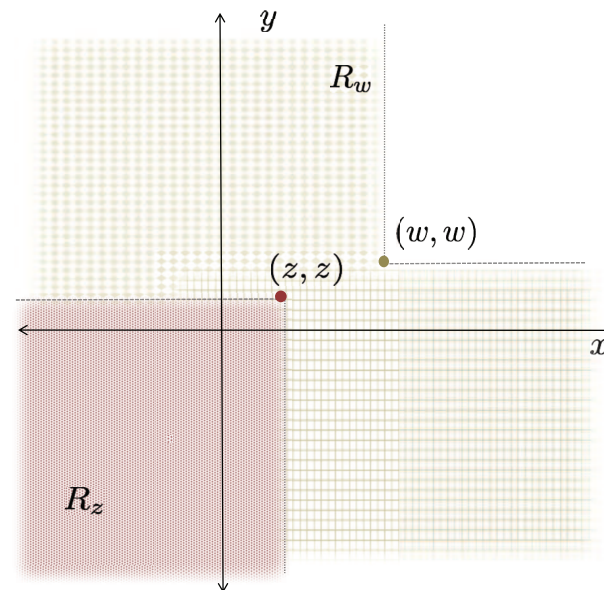
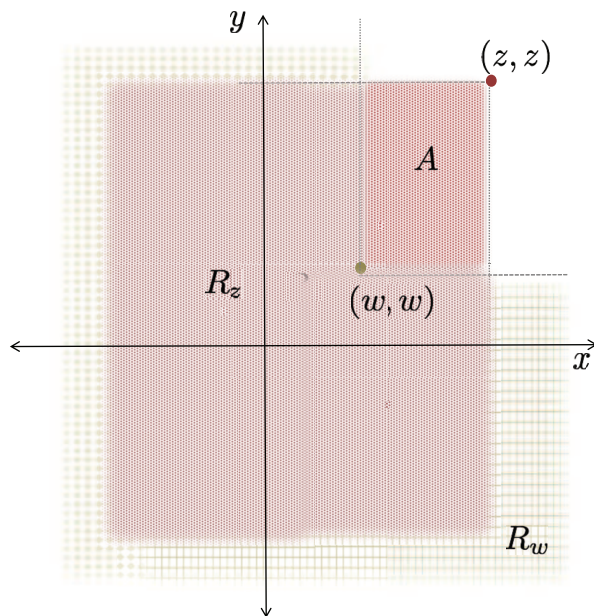
► R_z : $\{(x, y) : \max(\mathbf{x}, \mathbf{y}) \leq \mathbf{z} : \mathbf{x} \leq \mathbf{z} \text{ and } \mathbf{y} \leq \mathbf{z}\}$



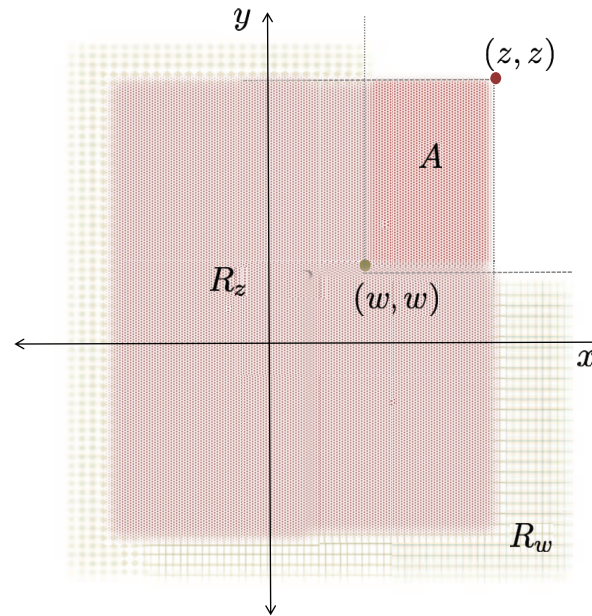
Two possible cases:

(a) if $z > w$, intersection $R_{w,z} = R_z - A$

(b) if $w > z$, intersection $R_{w,z} = R_z$

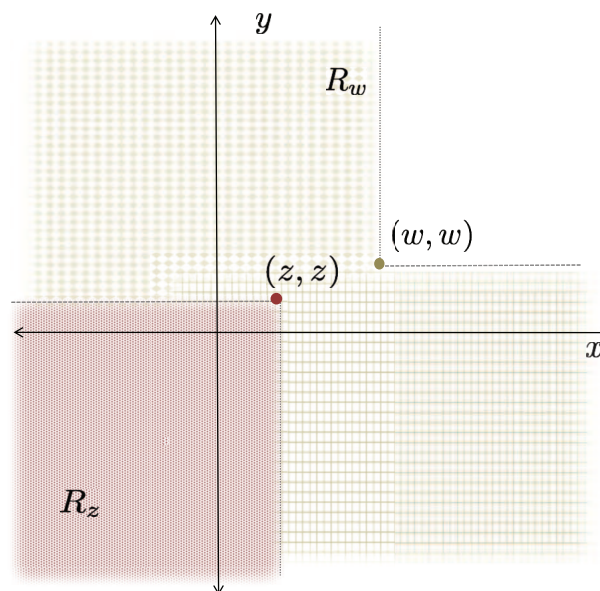


For (a): $z > w$, intersection $R_{w,z} = R_z - A$



$$\begin{aligned}
 \text{Joint CDF : } F_{WZ}(w, z) &= F_{XY}(z, z) - \Pr((x, y) \in A) \\
 &= F_{XY}(z, z) - \left[F_{XY}(z, z) + F_{XY}(w, w) - F_{XY}(w, z) - F_{XY}(z, w) \right] \\
 &= F_{XY}(w, z) + F_{XY}(z, w) - F_{XY}(w, w) \\
 \text{Joint PDF: } f_{WZ}(w, z) &= \frac{\partial^2}{\partial z \partial w} F_{WZ}(w, z) = f_{XY}(w, z) + f_{XY}(z, w)
 \end{aligned}$$

For (b): $w > z$, intersection $R_{w,z} = R_z$



Joint CDF: $F_{WZ}(w, z) = F_{XY}(z, z)$

Joint PDF

$$f_{WZ}(w, z) = \frac{\partial^2}{\partial z \partial w} F_{WZ}(w, z) = \frac{\partial^2}{\partial z \partial w} F_{XY}(z, z) = \frac{\partial}{\partial w} f_{XY}(z, z) = 0$$

Combining both cases, for transformations $W = \min(X, Y)$ and $Z = \max(X, Y)$

Joint CDF:

$$F_{WZ}(w, z) = \begin{cases} F_{XY}(w, z) + F_{XY}(z, w) - F_{XY}(w, w), & z > w \\ F_{XY}(z, z), & z < w \end{cases}$$

Joint PDF:

$$f_{WZ}(w, z) = \begin{cases} f_{XY}(w, z) + f_{XY}(z, w), & z > w \\ 0, & z < w \end{cases}$$